

解

计算 2×2 的数字图像 $f(0,0) = 2+i$, $f(0,1) = 3+2i$.

$f(1,0) = 2-2i$, $f(1,1) = 1+3i$ 的傅利叶相位谱。

请给出这道题的求解过程。

$$f = \begin{bmatrix} 2+i & 3+2i \\ 2-2i & 1+3i \end{bmatrix}, \text{尺寸 } M=2, N=2.$$

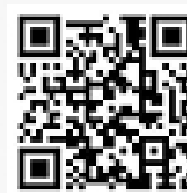
$$\bar{F}(u,v) = \sum_{x=0}^1 \sum_{y=0}^1 f(x,y) (-1)^{ux+vy}.$$

$$\begin{aligned} \bar{F}(0,0) &= f(0,0) + f(0,1) + f(1,0) + f(1,1) \\ &= (2+i) + (3+2i) + (2-2i) + (1+3i) \\ &= 8+4i. \end{aligned}$$

$$\begin{aligned} \bar{F}(0,1) &= f(0,0) - f(0,1) + f(1,0) - f(1,1) \\ &= (2+i) - (3+2i) + (2-2i) - (1+3i) \\ &= -6i \end{aligned}$$

$$\begin{aligned} \bar{F}(1,0) &= f(0,0) + f(0,1) - f(1,0) - f(1,1) \\ &= (2+i) + (3+2i) - (2-2i) - (1+3i) \\ &= 2+2i \end{aligned}$$

$$\begin{aligned} \bar{F}(1,1) &= f(0,0) - f(0,1) - f(1,0) + f(1,1) \\ &= (2+i) - (3+2i) - (2-2i) + (1+3i) \\ &= -2+4i. \end{aligned}$$



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$$\phi(u, v) = \arctan \left(\frac{\operatorname{Im} [f(u, v)]}{\operatorname{Re} [f(u, v)]} \right).$$

$$\phi(0, 0) = \arctan \left(\frac{1}{2} \right)$$

$$\phi(0, 1) = \arctan \left(-\frac{1}{2} \right) = -\frac{\pi}{2}$$

$$\phi(1, 0) = \arctan(1) = \frac{\pi}{4}$$

$$\phi(1, 1) = \arctan \left(-\frac{1}{2} \right) = \pi - \arctan(2).$$

$$\text{即 } \Phi(u, v) = \begin{bmatrix} \arctan \left(\frac{1}{2} \right), & -\frac{\pi}{2} \\ \frac{\pi}{4}, & \pi - \arctan(2) \end{bmatrix}.$$

